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Grade 11 Chemical Bonding and Hybridisation — Issue #36 Candidate B: Construction

Contents and concept route

Concept Book

Chemical Bonding and Hybridisation

Concept route

1. Cumulated pi bonding and allene stereochemical orthogonality
 1. Assign local bases before deducing the relation between terminal planes.
 2. Explain the terminal-plane constraint using overlap geometry, not an invalid occupation claim.
2. Delocalised pi systems and stationary resonance hybrids
 1. Separate supplied bond equivalence from a permanently chosen Lewis double bond.
 2. Translate formal contributors into one sigma framework and a complete pi-space ledger.
 3. Include the terminal-oxygen contributions when describing carbonate pi occupancy.

3. Local hybridisation models, rotational overlap symmetry, and conjugated pathways

1. Evaluate bond equivalence as a criterion rather than as a conclusion.

2. Assign each carbon locally and count sigma/pi components across the whole structure.

3. Separate persistence of the axial sigma connection from claims about all electronic interactions.

4. Trace the continuous aligned p pathway rather than counting all double bonds.

5. Synthesize a local procedure that also checks inter-centre compatibility.

Sections

Assign local bases before deducing the relation between terminal planes.

Explain the terminal-plane constraint using overlap geometry, not an invalid occupation claim.

Cumulated pi bonding and allene stereochemical orthogonality

Subject: Chemistry

Topic: Chemical Bonding

Concept: Cumulated pi bonding and allene stereochemical orthogonality

Concept difficulty: Hard concept

1

Assign local bases before deducing the relation between terminal planes.

Hard concept

Section 1 of 10

Next →

Conditions for this construction

- Can count local electron domains and identify sp vs sp^2 geometries.

Why this section matters for the question set

This unit builds toward

- Q3 · D3

The move learners miss: Allocate the local valence bases. [Try it in Core2](#)

1 Count local sigma-neighbour directions in allene.

Why valid: Each terminal carbon has two H plus the central C; the central carbon has two terminal-C neighbours.

Result: Terminals: three directions; centre: two.

2 The step the hard question turns on Allocate the local valence bases.

Why valid: Terminal sp^2 leaves one p each; central sp leaves two mutually perpendicular p functions.

Result: sp^2 - sp - sp^2 along the carbon chain.

3 Match the two separate central p directions to terminal p functions.

Why valid: Each localized pi component needs suitable lateral alignment with its terminal p.

Result: Two orthogonal pi planes.

4 Deduce and check terminal CH₂ planes.

Why valid: Each terminal sigma plane is perpendicular to its participating p; looking down the chain makes the relation explicit.

Result: Terminal CH₂ planes are perpendicular in this introductory model; rigid rotation preserves that relation.

Allene connectivity: $\text{H}_2\text{C}=\text{C}:$

Construct directions; do not

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

The key step

Allene uses terminal sp^2 , central sp accounting. The two central perpendicular p directions support perpendicular terminal CH_2 planes in the introductory model.

Formula reference · meaning and conditions

—

Equations and validity

Equation	What it tells you	Where can you
Two perpendicular central p directions; each aligns with its terminal partner.	Allene central sp accounting retains two perpendicular p functions. Terminal sigma planes are perpendicular to their respective p functions. Do not claim the first p orbital is filled by one bond and forbidden from multicentre participation by Pauli.	• •

Worked explanation

Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?

Follow the operation, its justification and its result.

1. A central sp carbon retains two perpendicular p directions.
2. Ethene uses one aligned pi pair; allene central carbon uses two perpendicular p directions. Repeating a single planar p direction for both exposes the mapping error.
3. Allene uses terminal sp², central sp accounting. The two central perpendicular p directions support perpendicular terminal CH₂ planes in the introductory model.

Result

Ethene uses one aligned pi pair; allene central carbon uses two perpendicular p directions. Repeating a single planar p direction for both exposes the mapping

error.

Check

Changed check: looking down $C=C=C$, terminal CH_2 planes project at right angles; rotating the whole model leaves that relation unchanged.

Mistake clinic • diagnose and repair

A tempting wrong path

- All double-bonded fragments must share one plane.
- Pauli exclusion forbids one atomic orbital from participating in any multicentre description.

Diagnose

- Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?
- Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

Repair

- Allene uses terminal sp^2 , central sp accounting. The two central perpendicular p directions support perpendicular terminal CH_2 planes in the introductory model.
- Allene central sp accounting retains two perpendicular p functions. Terminal sigma planes are perpendicular to their respective p functions. Do not claim the first p orbital is filled by one bond and forbidden from multicentre participation by Pauli.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: looking down $C=C=C$, terminal CH_2 planes project at right angles; rotating the whole model leaves that relation unchanged.

2 • APPLY

Build the allene end-on projection from the two p directions.

3 · CONNECT

A central sp carbon retains two perpendicular p directions.

2

Explain the terminal-plane constraint using overlap geometry, not an invalid occupation claim.

Hard concept

Section 2 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q4 · D3**

The move learners miss: Reallocate the central carbon for allene.

[Try it in Core2](#)

1 Construct the ethene baseline.

Why valid: Each carbon uses three planar sigma directions and one residual perpendicular p.

Result: One pair of parallel p functions; a coplanar sigma framework.

2 The step the hard question turns on **Reallocate the central carbon for allene.**

Why valid: Its two sigma-neighbour directions use sp, leaving two perpendicular p functions.

Result: Two independent central p directions rather than ethene's single p direction.

3 Match each terminal p to its corresponding central direction.

Why valid: The terminal sigma plane is perpendicular to its p, so distinct orthogonal p alignments constrain distinct planes.

Result: Perpendicular terminal CH₂ planes.

4 Reject both a copied planar sketch and the Pauli explanation.

Why valid: The supplied localized construction requires orthogonal directions; Pauli limits state occupancy, not all multicentre participation of one basis function.

Result: The plane relation follows orbital alignment in this model, not a universal one-bond-per-orbital prohibition.

Allene connectivity: $\text{H}_2\text{C}=\text{C}:$ Construct directions; do not

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Worked explanation

Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

Follow the operation, its justification and its result.

1. Atomic basis functions participate in molecular combinations; Pauli exclusion is an electron-state rule.
2. No. Allene uses distinct perpendicular local p directions in its model; nitrate is multicentre. A persistent forbidden-reuse claim identifies the faulty warrant.
3. Allene central sp accounting retains two perpendicular p functions. Terminal sigma planes are perpendicular to their respective p functions. Do not claim

the first p orbital is filled by one bond and forbidden from multicentre participation by Pauli.

Result

No. Allene uses distinct perpendicular local p directions in its model; nitrate is multicentre. A persistent forbidden-reuse claim identifies the faulty warrant.

Check

Changed check: a delocalised nitrate p function participates in a multicentre description, falsifying the claimed one-AO-one-bond prohibition.

Mistake clinic • diagnose and repair

A tempting wrong path

- All double-bonded fragments must share one plane.
- Pauli exclusion forbids one atomic orbital from participating in any multicentre description.

Diagnose

- Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?
- Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

Repair

- Allene uses terminal sp^2 , central sp accounting. The two central perpendicular p directions support perpendicular terminal CH_2 planes in the introductory model.
- Allene central sp accounting retains two perpendicular p functions. Terminal sigma planes are perpendicular to their respective p functions. Do not claim the first p orbital is filled by one bond and forbidden from multicentre participation by Pauli.

1-2-3 QUICK CHECK

■

1 • CHECK

Changed check: a delocalised nitrate p function participates in a multicentre description, falsifying the claimed one-AO-one-bond prohibition.

2 • APPLY

Explain allene perpendicularity without using the phrase orbital already occupied.

3 • CONNECT

Atomic basis functions participate in molecular combinations; Pauli exclusion is an electron-state rule.

Question-specific diagnosis and repair

Repair for Q3

Test the rule on a changed case

Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?

Your prediction and reason

Repair for Q4

Test the rule on a changed case

Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

Your prediction and reason

Now you do one

Number carbons left-to-right in $\text{CH}_3\text{-CH}=\text{C}=\text{CH}_2$. Assign each local introductory basis and the terminal-plane relationship for the cumulated double-bond triad.

Model answer

Written-order labels: C1 sp^3 , C2 sp^2 , C3 sp , C4 sp^2 . The sigma planes at the two terminal carbons of the cumulated triad are perpendicular in the introductory model.

1. Written-order labels: C1 sp^3 , C2 sp^2 , C3 sp , C4 sp^2 . The sigma planes at the two terminal carbons of the cumulated triad are perpendicular in the introductory model.

Practice this concept in Core2

Q3

Q4

Sections

Separate supplied bond equivalence from a permanently chosen Lewis double bond.

Translate formal contributors into one sigma framework and a complete pi-space ledger.

Include the terminal-oxygen contributions when describing carbonate pi occupancy.

Delocalised pi systems and stationary resonance hybrids

Subject: Chemistry

Topic: Chemical Bonding

Concept: Delocalised pi systems and stationary resonance hybrids

Concept difficulty: Hard concept

1

Separate supplied bond equivalence from a permanently chosen Lewis double bond.

Hard concept

Section 3 of 10

← Previous

Next →

Conditions for this construction

- Can draw valid Lewis structures with resonance contributors.

Why this section matters for the question set

This unit builds toward

- Q1 · D2

The move learners miss: Choose one delocalised pi account for all contributors. [Try it in Core2](#)

1 Compare a fixed localized nitrate picture with the supplied equivalence.

Why valid: A permanently unique N=O position distinguishes one bond; the observation instead supplies three equivalent N-O bonds.

Result: The fixed single/double-bond assignment is inadequate.

2 The step the hard question turns on Choose one delocalised pi account for all contributors.

Why valid: The contributors are alternative bookkeeping descriptions of one electronic state, not switching structures.

Result: One delocalised pi system spans the N-O framework.

3 Specify nitrogen sigma accounting independently.

Why valid: Three planar local sigma directions support the introductory sp² framework; equivalence alone is not its criterion.

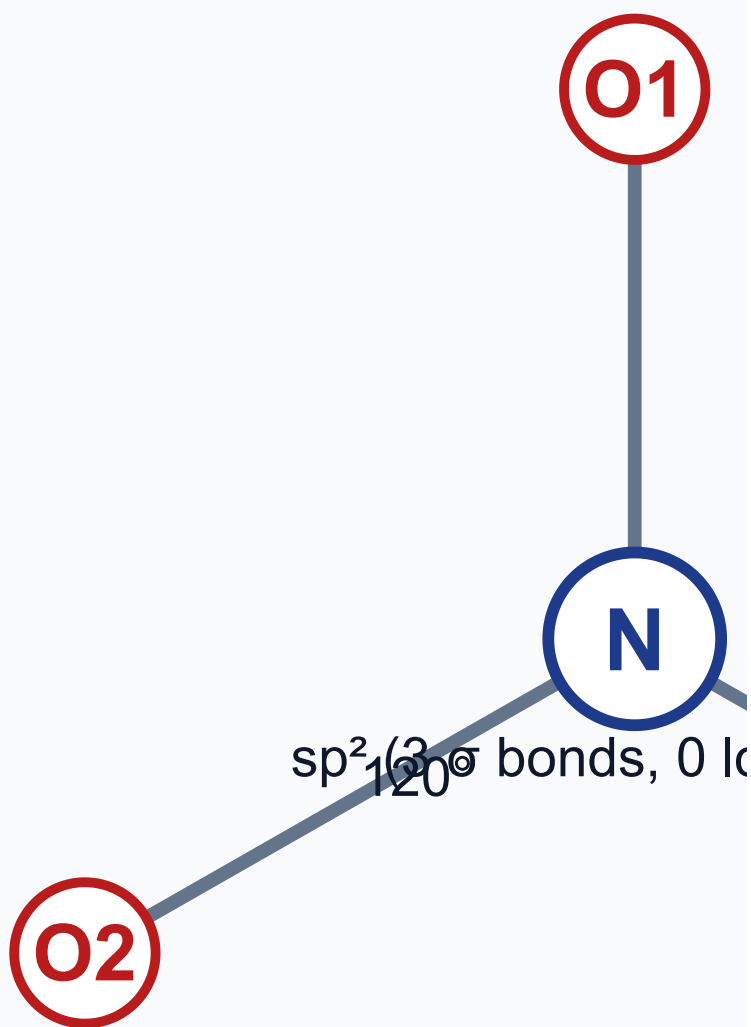
Result: Three N sigma directions; residual perpendicular p participates in the pi space.

4 Test the equivalence-implies-sp² inference on NH₃.

Why valid: NH₃ can have three equivalent N-H bonds while having four local domains.

Result: Bond equivalence does not uniquely select an sp² description.

Stage 1: Trigonal Planar Sigma Framework



Identify sp² nitrogen and 120° bond angles.

Stage 1 of 3

Represent planar sigma framework and continuous 4-centre delocalised pi system in nitrate.

The key step

Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.

Formula reference • meaning and conditions

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Equations and validity

Equation	What it tells you	When you use it
Four perpendicular p basis functions; six pi-space electrons.	Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge $-2/3$ is contributor bookkeeping, not a uniquely measured atomic partial charge.	• Planar trigonal geometry with 1 point symmetry • Unhybridized 2pz orbitals mutually perpendicular

Worked explanation

Can equivalent bonds at CO₂ carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.

Follow the operation, its justification and its result.

1. Equivalent contributors are bookkeeping alternatives for one delocalised description.
2. No: two versus three directions in these local models.
3. Planar equivalent nitrate bonds reject one permanently unique N=O position. N sigma framework is locally sp²; equivalence alone is not the

hybridisation criterion.

Result

No: two versus three directions in these local models.

Check

Changed check: supplied equivalent CO₂ bonds coexist with sp carbon; equivalence cannot alone imply sp².

Mistake clinic • diagnose and repair

A tempting wrong path

- Equivalent bonds are a permanently selected Lewis double bond.
- Resonance contributors are molecular structures switching in time; only the drawn double bond supplies pi electrons.
- A delocalised carbonate pi space contains only one pair; formal charge averages are measured partial charges.

Diagnose

- Can equivalent bonds at CO₂ carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.
- For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?
- How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Repair

- Planar equivalent nitrate bonds reject one permanently unique N=O position. N sigma framework is locally sp²; equivalence alone is not the hybridisation criterion.
- Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.

- Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge $-2/3$ is contributor bookkeeping, not a uniquely measured atomic partial charge.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: supplied equivalent CO₂ bonds coexist with sp carbon; equivalence cannot alone imply sp².

2 • APPLY

Apply the distinction to supplied equivalent carbonate bonds.

3 • CONNECT

Equivalent contributors are bookkeeping alternatives for one delocalised description.

2

Translate formal contributors into one sigma framework and a complete pi-space ledger.

Hard concept

Section 4 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q2 • D3

The move learners miss: Build a perpendicular p basis across all four centres. [Try it in Core2](#)

1 Preserve the nitrate nuclei and sigma connectivity.

Why valid: Every supplied contributor has the same N centre and three N-O connections.

Result: Three coplanar N-O sigma directions.

2 The step the hard question turns on Build a perpendicular p basis across all four centres.

Why valid: A delocalised qualitative diagram combines the central N and three O p functions.

Result: Four perpendicular p functions spanning one pi space.

3 Count pi-space electrons from one contributor.

Why valid: One N=O pi pair and p-oriented pairs on the two singly bonded O sites supply three pairs.

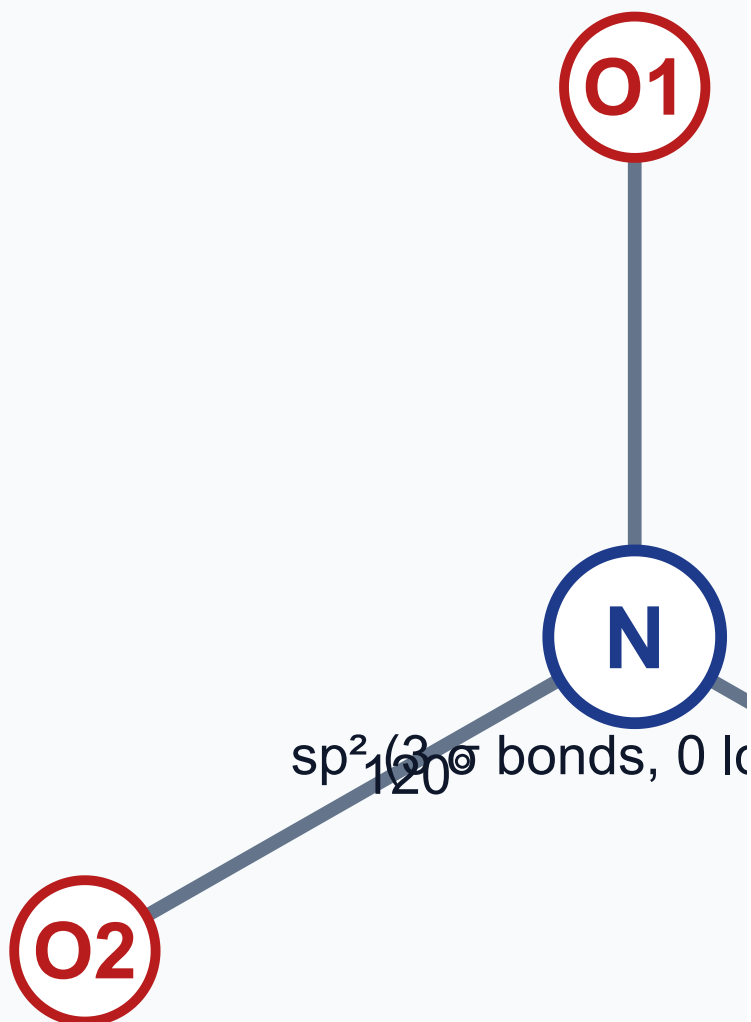
Result: Six pi-space electrons; the Lewis double bond alone is not the complete pi inventory.

4 Check contributor invariants rather than a switching animation.

Why valid: Changing the drawn N=O position changes bookkeeping, not nuclei, total charge or electron number.

Result: One planar equivalent-bond state; no molecule oscillating between three structures.

Stage 1: Trigonal Planar Sigma Framework



Identify sp^2 nitrogen and 120° bond angles.

Stage 1 of 3

Represent planar sigma framework and continuous 4-centre delocalised pi system in nitrate.

Worked explanation

For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?

Follow the operation, its justification and its result.

1. A formal Lewis double bond does not identify the whole delocalised occupancy.
2. Six pi-space electrons in each; no switching of nuclear structures.
3. Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.

Result

Six pi-space electrons in each; no switching of nuclear structures.

Check

Changed check: 24 total valence electrons = 18 in sigma/in-plane lone-pair bookkeeping plus six in the perpendicular pi space.

Mistake clinic · diagnose and repair

A tempting wrong path

- Equivalent bonds are a permanently selected Lewis double bond.
- Resonance contributors are molecular structures switching in time; only the drawn double bond supplies pi electrons.
- A delocalised carbonate pi space contains only one pair; formal charge averages are measured partial charges.

Diagnose

- Can equivalent bonds at CO₂ carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.
- For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?
- How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Repair

- Planar equivalent nitrate bonds reject one permanently unique N=O position. N sigma framework is locally sp²; equivalence alone is not the hybridisation criterion.
- Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.
- Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge -2/3 is contributor bookkeeping, not a uniquely measured atomic partial charge.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: 24 total valence electrons = 18 in sigma/in-plane lone-pair bookkeeping plus six in the perpendicular pi space.

2 • APPLY

Translate carbonate contributors into a continuous p-space description.

3 • CONNECT

A formal Lewis double bond does not identify the whole delocalised occupancy.

3

Include the terminal-oxygen contributions when describing carbonate pi occupancy.

Hard concept

Section 5 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q7 · D3

The move learners miss: Construct the delocalised pi basis and electron ledger. [Try it in Core2](#)

1 Preserve carbonate connectivity across contributors.

Why valid: All contributors use one C centre and three O neighbours with total charge -2.

Result: Three planar C-O sigma directions.

2 The step the hard question turns on Construct the delocalised pi basis and electron ledger.

Why valid: Central C plus three O p functions span the pi space; a contributor has one pi pair plus p-oriented pairs at two singly bonded O sites.

Result: Four p functions and six pi-space electrons.

3 Interpret contributor charge bookkeeping.

Why valid: Three equivalent contributors average the two formal negative O charges across three sites.

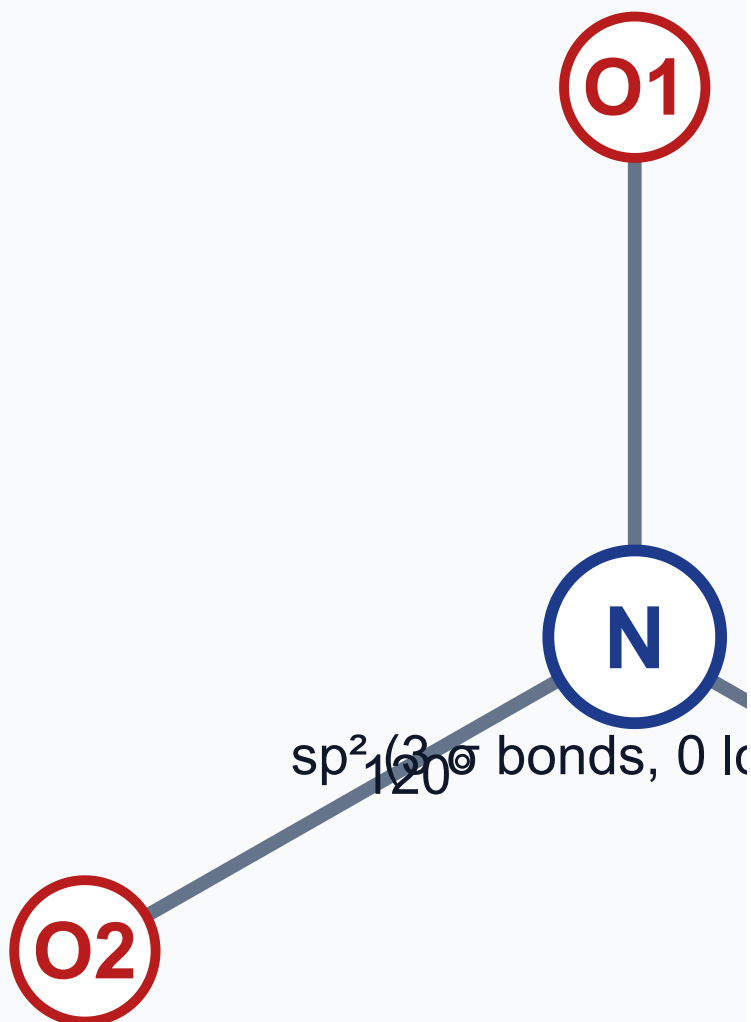
Result: Average formal charge $-2/3$ per O; this is not a uniquely measured partial charge.

4 Check why one fixed contributor is incomplete.

Why valid: A permanently selected C=O position conflicts with the supplied three equivalent C-O bonds.

Result: Use one delocalised description, preserving nuclei, total charge and electron count.

Stage 1: Trigonal Planar Sigma Framework



Identify sp^2 nitrogen and 120° bond angles.

Stage 1 of 3

Represent planar sigma framework and continuous 4-centre delocalised pi system in nitrate.

Worked explanation

How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Follow the operation, its justification and its result.

1. Formal charge averages and physical partial charges have different meanings.
2. Six; no.
3. Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge $-2/3$ is contributor bookkeeping, not a uniquely measured atomic partial charge.

Result

Six; no.

Check

Independent ledger: CO_3^{2-} has 24 valence electrons; three sigma pairs and six in-plane O lone pairs use eighteen, leaving six in the pi space.

Mistake clinic · diagnose and repair

A tempting wrong path

- Equivalent bonds are a permanently selected Lewis double bond.
- Resonance contributors are molecular structures switching in time; only the drawn double bond supplies pi electrons.
- A delocalised carbonate pi space contains only one pair; formal charge averages are measured partial charges.

Diagnose

- Can equivalent bonds at CO_2 carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.
- For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?
- How many carbonate pi-space electrons arise from one $\text{C}=\text{O}$ pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Repair

- Planar equivalent nitrate bonds reject one permanently unique N=O position. N sigma framework is locally sp^2 ; equivalence alone is not the hybridisation criterion.
- Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.
- Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge $-2/3$ is contributor bookkeeping, not a uniquely measured atomic partial charge.

1-2-3 QUICK CHECK

1 • CHECK

Independent ledger: CO_3^{2-} has 24 valence electrons; three sigma pairs and six in-plane O lone pairs use eighteen, leaving six in the pi space.

2 • APPLY

Explain why equal bond lengths do not uniquely determine atomic partial charges.

3 • CONNECT

Formal charge averages and physical partial charges have different meanings.

Question-specific diagnosis and repair

Repair for Q1

Test the rule on a changed case

Can equivalent bonds at CO_2 carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.

Your prediction and reason

Repair for Q2

Test the rule on a changed case

For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?

Your prediction and reason

Repair for Q7

Test the rule on a changed case

How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Your prediction and reason

Now you do one

A hypothetical planar oxyanion is supplied with equivalent bonds and several formal contributors. Does drawing several contributors mean the nuclei switch between them in time? State a further claim that equivalence alone cannot prove.

Model answer

No. Contributors describe one delocalised model, not alternating nuclear structures. Equivalence alone does not uniquely measure an orbital

decomposition or physical partial charges.

1. No. Contributors describe one delocalised model, not alternating nuclear structures. Equivalence alone does not uniquely measure an orbital decomposition or physical partial charges.

Practice this concept in Core2

Q1

Q2

Q7

Sections

—

Evaluate bond equivalence as a criterion rather than as a conclusion.

Assign each carbon locally and count sigma/pi components across the whole structure.

Separate persistence of the axial sigma connection from claims about all electronic interactions.

Trace the continuous aligned p pathway rather than counting all double bonds.

Synthesize a local procedure that also checks inter-centre compatibility.

Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

1

Evaluate bond equivalence as a criterion rather than as a conclusion.

Hard concept

Section 6 of 10

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Conditions for this construction

- Can assign steric numbers to central atoms and count sigma/pi bonds.

Why this section matters for the question set

This unit builds toward

- **Q5** · D3 · the hardest question in this set
The move learners miss: Use CO₂ or NH₃ as a counterexample.
[Try it in Core2](#)

1 List the supplied local inventories beside equivalence.

Why valid: CO₂ C has two domains; nitrate N three; NH₃ N four although each case has equivalent central bonds.

Result: Two, three and four local totals with equivalence held true.

2 The step the hard question turns on Use CO₂ or NH₃ as a counterexample.

Why valid: One case satisfying the premise but failing the claimed sp² consequence refutes a universal implication.

Result: CO₂ uses introductory sp; NH₃ sp³-like; the claim is false.

3 Separate model labels from observable bond equality.

Why valid: Domain count and orbital allocation are different layers from measured equivalence.

Result: Only nitrate's three-direction planar framework supports the local sp² description here.

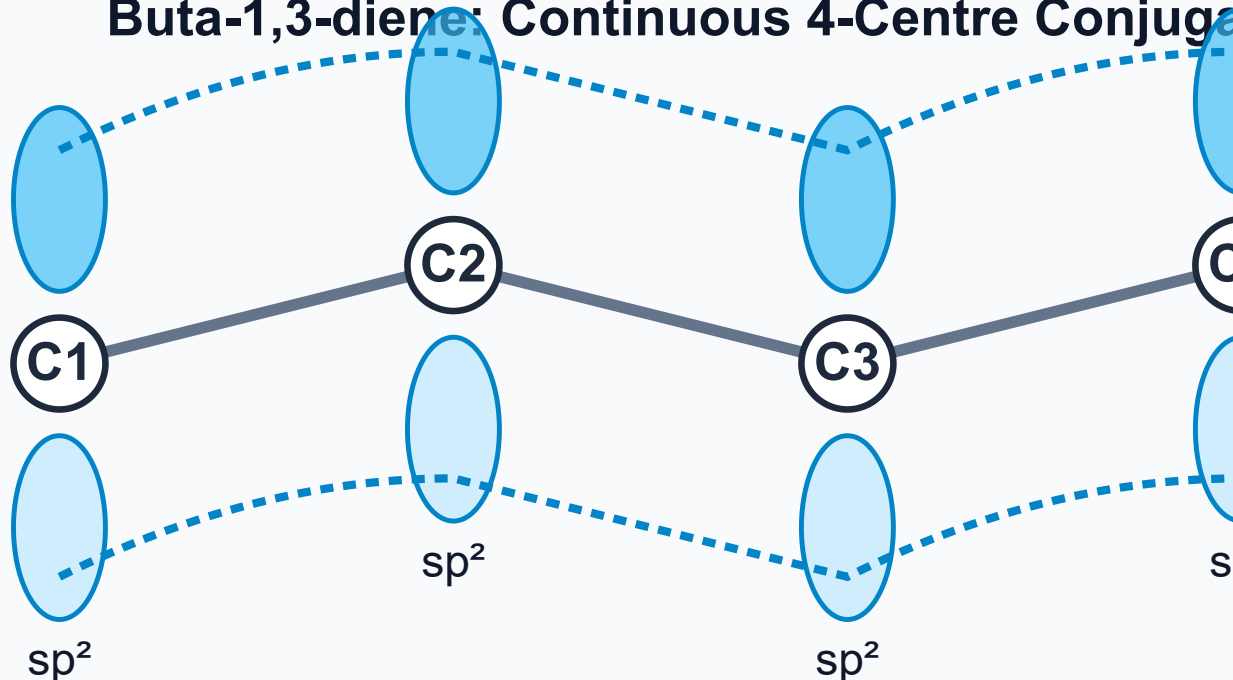
4 Change to CH₄ as an independent test.

Why valid: Equivalent C-H bonds coexist with four local directions and an

sp³ model.

Result: The rejected equivalence-only rule fails a fresh fourth case.

Buta-1,3-diene: Continuous 4-Centre Conjugation



Trace continuous sequence of four parallel p_z orbitals.

Stage 1 of 3

Contrast continuous 4-centre conjugation in butadiene with interruption by sp³ carbon in pentadiene.

The key step

Equivalent bonds do not imply sp²: CO₂ carbon is sp, nitrate N sp², and NH₃ N sp³-like in their introductory models.

Formula reference · meaning and conditions

Equations and validity

Equation	What it tells you	When you use it
SN = sigma bonds + lone pairs; SN=2 → sp	Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring	Mul bon

Equation	What it tells you	When you use it
(180°) , $SN=3 \rightarrow$ sp^2 (120°), $SN=4$ $\rightarrow sp^3$ (109.5°)	compatibility, then qualify applicability. Propyne/allene classifications are atom-local.	could be one electron domain • Not a bond lone pair could be a stereo number

Worked explanation

Apply the proposed equality-implies- sp^2 rule to CH_4 and CO_2 , then compare it with a domain inventory.

Follow the operation, its justification and its result.

1. Equivalent bond properties and local orbital-basis selection are separate statements.
2. Four versus two domains refute the shared label. Repeating sp^2 solely from equivalence identifies criterion confusion.
3. Equivalent bonds do not imply sp^2 : CO_2 carbon is sp , nitrate N sp^2 , and NH_3 N sp^3 -like in their introductory models.

Result

Four versus two domains refute the shared label. Repeating sp^2 solely from equivalence identifies criterion confusion.

Check

Changed check: CH₄ has equivalent C-H bonds but four carbon directions, contradicting equality-implies-sp².

Mistake clinic • diagnose and repair

A tempting wrong path

- Bond equivalence by itself implies sp².
- Each triple-bond line is a sigma bond.
- A fixed central sigma axis means every molecular interaction is unchanged.
- Spatial closeness repairs an interrupted p pathway.
- A whole molecule has one hybridisation label.

Diagnose

- Apply the proposed equality-implies-sp² rule to CH₄ and CO₂, then compare it with a domain inventory.
- Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?
- During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.
- Compare CH₂=CH-CH=CH₂ with CH₂=CH-CH₂-CH=CH₂. Does spatial closeness alone repair the missing p pathway?
- Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Repair

- Equivalent bonds do not imply sp²: CO₂ carbon is sp, nitrate N sp², and NH₃ N sp³-like in their introductory models.
- Propyne CH₃-C≡CH has carbon labels sp³, sp, sp; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.
- Ethane axial C-C sigma bonding can persist through torsion; ethene pi alignment changes under twist. Other interactions, including hyperconjugation, can change; an exclusively steric total-energy account is unjustified.

- Butadiene has four contiguous p-bearing centres; a central saturated CH₂ in separated pentadiene interrupts this introductory pathway. This is not a proof of zero interaction of every quantum-mechanical kind.
- Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: CH₄ has equivalent C-H bonds but four carbon directions, contradicting equality-implies-sp².

2 • APPLY

Write a qualified hybridisation procedure whose inputs exclude bond equivalence as a sufficient test.

3 • CONNECT

Equivalent bond properties and local orbital-basis selection are separate statements.

2

Assign each carbon locally and count sigma/pi components across the whole structure.

Hard concept

Section 7 of 10

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Next →

Why this section matters for the question set

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This unit builds toward

- Q6 · D2

The move learners miss: Count sigma components by atom connectivity.

[Try it in Core2](#)

1 Classify each carbon from its own neighbours in propyne.

Why valid: CH₃ has four directions; both triple-bond carbons have two

sigma-neighbour directions.

Result: C labels in order: sp^3 , sp , sp .

2 The step the hard question turns on Count sigma components by atom connectivity.

Why valid: Each connected atom pair has one sigma component regardless of bond order.

Result: Four C-H plus two C-C sigma components = six.

3 Count the extra components of the triple bond.

Why valid: A triple bond contains one sigma and two pi components in the introductory account.

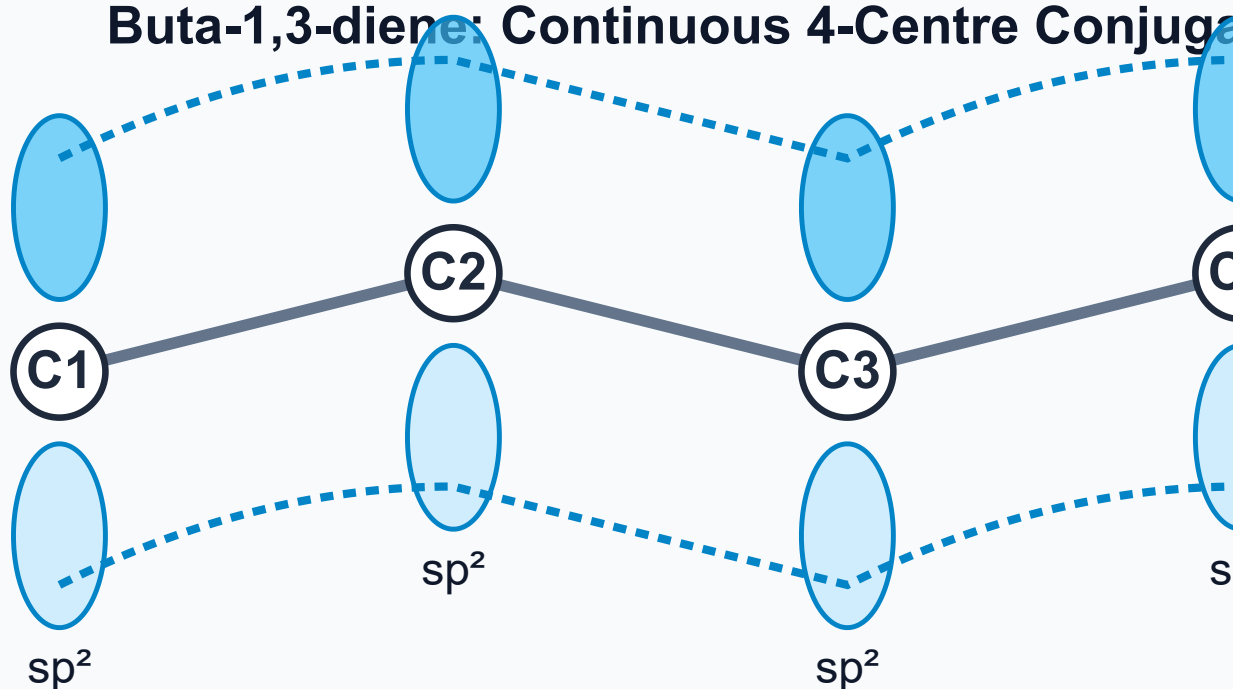
Result: Two pi bonds in the whole molecule.

4 Reconcile local and whole-molecule inventories.

Why valid: The triple bond connects one neighbour at either end even though it has three components.

Result: Six sigma/two pi globally; two local directions at each sp carbon, with no contradiction.

Buta-1,3-diene: Continuous 4-Centre Conjugation



Trace continuous sequence of four parallel p_z orbitals.

Contrast continuous 4-centre conjugation in butadiene with interruption by sp^3 carbon in pentadiene.

Worked explanation

Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?

Follow the operation, its justification and its result.

1. A triple bond supplies one sigma plus two pi components.
2. Propyne six, ethyne three; multiplicity adds pi components.
3. Propyne $CH_3-C\equiv CH$ has carbon labels sp^3 , sp , sp ; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.

Result

Propyne six, ethyne three; multiplicity adds pi components.

Check

Independent electron ledger: C_3H_4 has 16 valence electrons; six sigma pairs plus two pi pairs account for sixteen.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Bond equivalence by itself implies sp^2 .
- Each triple-bond line is a sigma bond.
- A fixed central sigma axis means every molecular interaction is unchanged.
- Spatial closeness repairs an interrupted p pathway.
- A whole molecule has one hybridisation label.

Diagnose

- Apply the proposed equality-implies-sp² rule to CH₄ and CO₂, then compare it with a domain inventory.
- Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?
- During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.
- Compare CH₂=CH-CH=CH₂ with CH₂=CH-CH₂-CH=CH₂. Does spatial closeness alone repair the missing p pathway?
- Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Repair

- Equivalent bonds do not imply sp²: CO₂ carbon is sp, nitrate N sp², and NH₃ N sp³-like in their introductory models.
- Propyne CH₃-C≡CH has carbon labels sp³, sp, sp; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.
- Ethane axial C-C sigma bonding can persist through torsion; ethene pi alignment changes under twist. Other interactions, including hyperconjugation, can change; an exclusively steric total-energy account is unjustified.
- Butadiene has four contiguous p-bearing centres; a central saturated CH₂ in separated pentadiene interrupts this introductory pathway. This is not a proof of zero interaction of every quantum-mechanical kind.
- Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

1-2-3 QUICK CHECK

1 • CHECK

Independent electron ledger: C₃H₄ has 16 valence electrons; six sigma pairs plus two pi pairs account for sixteen.

2 • APPLY

Classify each carbon and count bond components in ethyne.

3 • CONNECT

A triple bond supplies one sigma plus two pi components.

3

Separate persistence of the axial sigma connection from claims about all electronic interactions.

Hard concept

Section 8 of 10

← Previous

Next →

Why this section matters for the question set

This unit builds toward

- Q8 · D3

The move learners miss: Identify the extra ethene pi alignment requirement.

[Try it in Core2](#)

1 Specify the axial C-C sigma reference in both molecules.

Why valid: Rotation about that axis can preserve the axial sigma overlap.

Result: A sigma connection can persist during torsion.

2 The step the hard question turns on Identify the extra ethene pi alignment requirement.

Why valid: Its parallel residual p functions overlap laterally; relative group twist changes that alignment.

Result: Ethene twist alters pi overlap; ethane has no corresponding C-C pi component.

3 Separate rigid rotation from relative torsion.

Why valid: Only relative orientation about the bond changes the participating p relation.

Result: Rotating the whole molecule does not reproduce twisting one group.

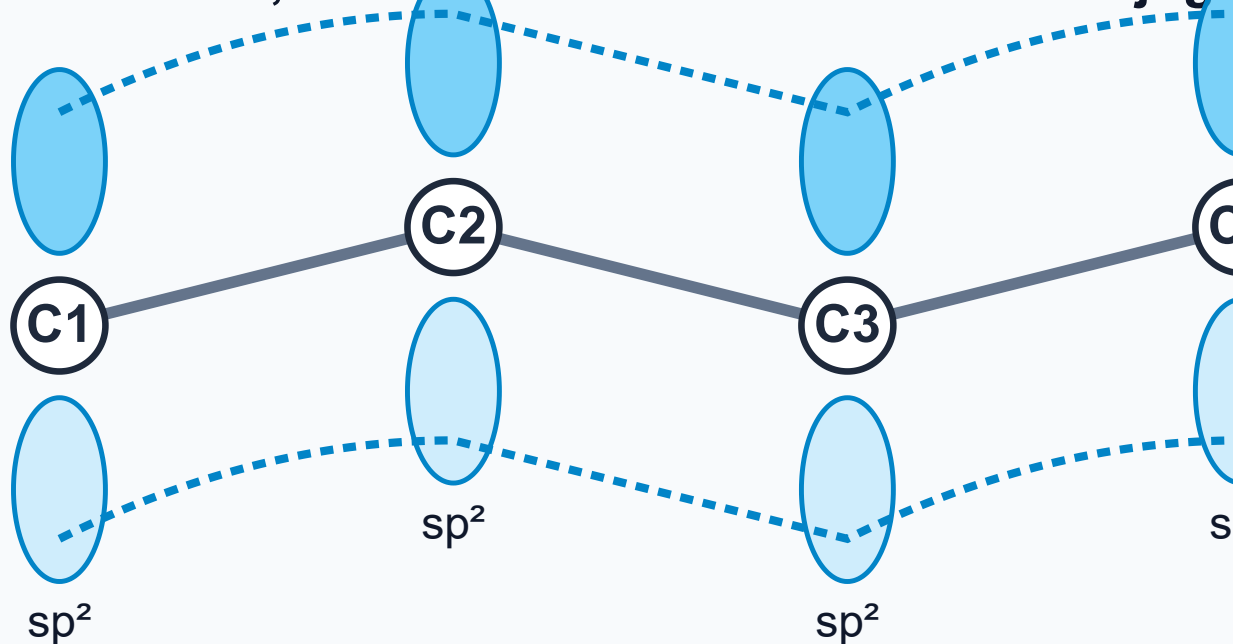
4 Bound the energy inference.

Why valid: Other interactions, including hyperconjugation, can change even

when the central sigma connection remains.

Result: No numerical barrier or exclusively steric total-energy law follows from the sketch.

Buta-1,3-diene: Continuous 4-Centre Conjugation



Trace continuous sequence of four parallel p_z orbitals.

Stage 1 of 3

Contrast continuous 4-centre conjugation in butadiene with interruption by sp³ carbon in pentadiene.

Worked explanation

During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.

Follow the operation, its justification and its result.

1. Relative p alignment affects pi overlap; one axial sketch is not a total-energy decomposition.
2. No; ethene relative p alignment also changes. A whole-molecule no-overlap-change claim reveals model overextension.

3. Ethane axial C-C sigma bonding can persist through torsion; ethene pi alignment changes under twist. Other interactions, including hyperconjugation, can change; an exclusively steric total-energy account is unjustified.

Result

No; ethene relative p alignment also changes. A whole-molecule no-overlap-change claim reveals model overextension.

Check

Changed check: a fixed central sigma direction does not hold every C-H/donor-acceptor orientation fixed.

Mistake clinic • diagnose and repair

A tempting wrong path

- Bond equivalence by itself implies sp^2 .
- Each triple-bond line is a sigma bond.
- A fixed central sigma axis means every molecular interaction is unchanged.
- Spatial closeness repairs an interrupted p pathway.
- A whole molecule has one hybridisation label.

Diagnose

- Apply the proposed equality-implies- sp^2 rule to CH_4 and CO_2 , then compare it with a domain inventory.
- Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?
- During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.
- Compare $CH_2=CH-CH=CH_2$ with $CH_2=CH-CH_2-CH=CH_2$. Does spatial closeness alone repair the missing p pathway?
- Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Repair

- Equivalent bonds do not imply sp^2 : CO_2 carbon is sp , nitrate N sp^2 , and NH_3 N sp^3 -like in their introductory models.
- Propyne $CH_3-C\equiv CH$ has carbon labels sp^3 , sp , sp ; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.
- Ethane axial C-C sigma bonding can persist through torsion; ethene pi alignment changes under twist. Other interactions, including hyperconjugation, can change; an exclusively steric total-energy account is unjustified.
- Butadiene has four contiguous p-bearing centres; a central saturated CH_2 in separated pentadiene interrupts this introductory pathway. This is not a proof of zero interaction of every quantum-mechanical kind.
- Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: a fixed central sigma direction does not hold every C-H/donor-acceptor orientation fixed.

2 • APPLY

Explain the sigma/pi contrast without assigning a universal torsional energy cause.

3 • CONNECT

Relative p alignment affects pi overlap; one axial sketch is not a total-energy decomposition.

4

Trace the continuous aligned p pathway rather than counting all double bonds.

Hard concept

Section 9 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q9 · D3**

The move learners miss: Trace the second molecule using the same local test. [Try it in Core2](#)

1 Trace consecutive p-bearing carbon sites in butadiene.

Why valid: An approximately planar conjugated chain supplies a p function at each of its four sp² sites.

Result: A four-site contiguous aligned p pathway.

2 **The step the hard question turns on** Trace the second molecule using the same local test.

Why valid: Its intervening saturated CH₂ uses localized sp³ accounting, with no ordinary residual p in this model.

Result: The central spacer interrupts the contiguous pathway.

3 Distinguish adjacency from spatial proximity.

Why valid: Moving separated fragments near each other does not remove the saturated node from their bonding path.

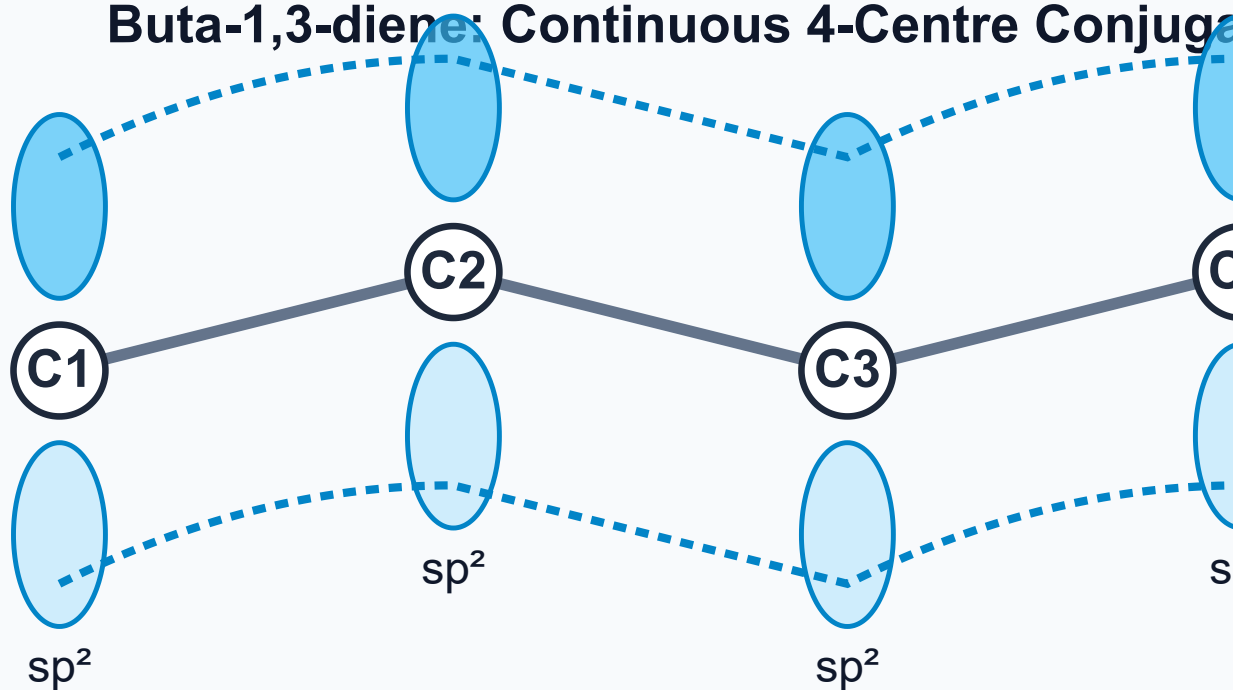
Result: The ordinary through-chain conjugation criterion remains unmet.

4 State the model boundary explicitly.

Why valid: Absence of the chosen contiguous p path does not exclude every possible quantum interaction.

Result: No absolute zero-interaction claim; reassess if the spacer becomes p-bearing and suitably aligned.

Buta-1,3-diene: Continuous 4-Centre Conjugation



Trace continuous sequence of four parallel p_z orbitals.

Stage 1 of 3

Contrast continuous 4-centre conjugation in butadiene with interruption by sp³ carbon in pentadiene.

Worked explanation

Compare CH₂=CH-CH=CH₂ with CH₂=CH-CH₂-CH=CH₂. Does spatial closeness alone repair the missing p pathway?

Follow the operation, its justification and its result.

1. An intervening saturated sp³ centre breaks the ordinary contiguous pi pathway in this model.
2. No; the first has a continuous path, the second an sp³ interruption.
3. Butadiene has four contiguous p-bearing centres; a central saturated CH₂ in separated pentadiene interrupts this introductory pathway. This is not a proof of zero interaction of every quantum-mechanical kind.

Result

No; the first has a continuous path, the second an sp³ interruption.

Check

Changed check: replacing the saturated spacer with an appropriate unsaturated p-bearing centre restores a contiguous path.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Bond equivalence by itself implies sp^2 .
- Each triple-bond line is a sigma bond.
- A fixed central sigma axis means every molecular interaction is unchanged.
- Spatial closeness repairs an interrupted p pathway.
- A whole molecule has one hybridisation label.

Diagnose

- Apply the proposed equality-implies- sp^2 rule to CH_4 and CO_2 , then compare it with a domain inventory.
- Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?
- During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.
- Compare $CH_2=CH-CH=CH_2$ with $CH_2=CH-CH_2-CH=CH_2$. Does spatial closeness alone repair the missing p pathway?
- Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Repair

- Equivalent bonds do not imply sp^2 : CO_2 carbon is sp , nitrate N sp^2 , and NH_3 N sp^3 -like in their introductory models.
- Propyne $CH_3-C\equiv CH$ has carbon labels sp^3 , sp , sp ; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.
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- Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: replacing the saturated spacer with an appropriate unsaturated p-bearing centre restores a contiguous path.

2 • APPLY

Specify a structural change that restores the pathway and a model limitation.

3 • CONNECT

An intervening saturated sp³ centre breaks the ordinary contiguous pi pathway in this model.

5

Synthesize a local procedure that also checks inter-centre compatibility.

Hard concept

Section 10 of 10

← Previous

Why this section matters for the question set

—

This unit builds toward

- **Q10** · D3

The move learners miss: Test the procedure on allene. [Try it in Core2](#)

1 Test one-label-per-molecule on propyne.

Why valid: Its carbon local inventories differ: four, two, two directions.

Result: sp³, sp, sp; a single whole-molecule label fails.

2 The step the hard question turns on Test the procedure on allene.

Why valid: The terminal and central inventories are three, two, three directions.

Result: sp^2 , sp , sp^2 ; two perpendicular central p directions must remain.

3 Build the replacement atom-local procedure.

Why valid: Count neighbours and lone pairs, conserve each local valence basis, then assign residual p roles.

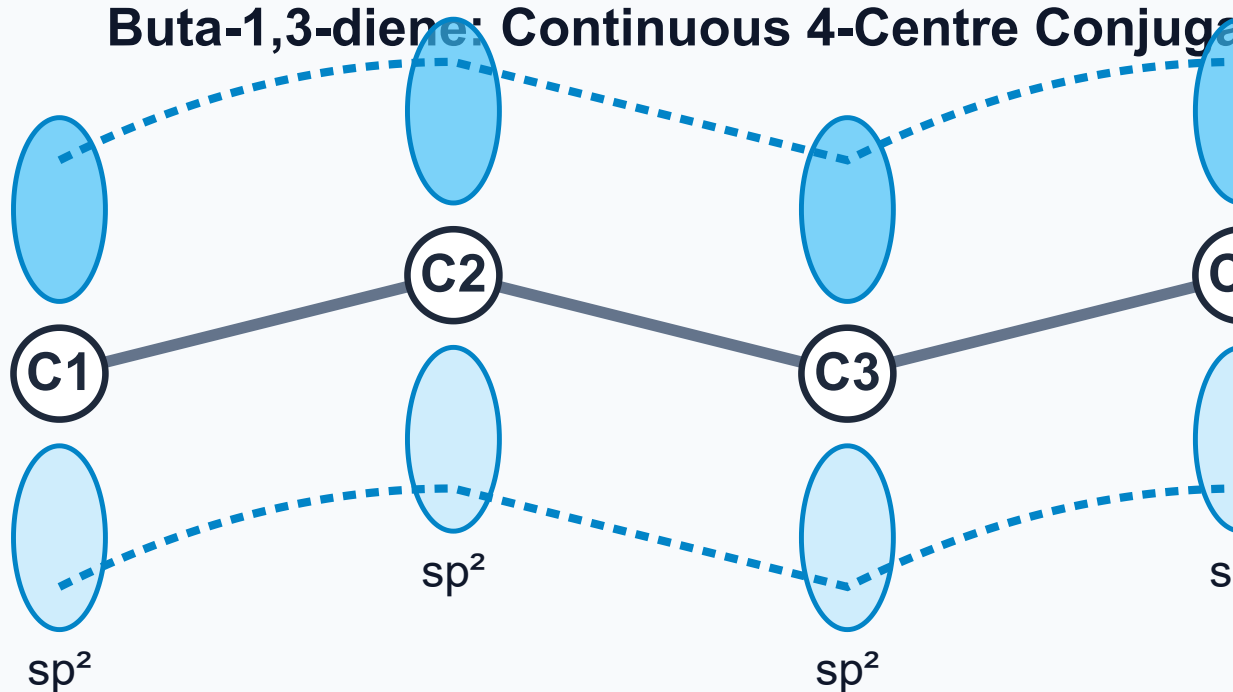
Result: A label and orbital ledger for each atom.

4 Check neighbouring orientation and applicability.

Why valid: Adjacent p functions must support the intended pi account; delocalisation or unusual occupancy can require a richer model.

Result: A qualified local framework with explicit compatibility and stop-for-more-evidence checks.

Buta-1,3-diene: Continuous 4-Centre Conjugation



Trace continuous sequence of four parallel p_z orbitals.

Stage 1 of 3

Contrast continuous 4-centre conjugation in butadiene with interruption by sp^3 carbon in pentadiene.

Worked explanation

Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Follow the operation, its justification and its result.

1. Local count, residual p directions and model applicability have different jobs.
2. No; the positions differ. Repeating one molecule-wide label identifies failure to localise the procedure.
3. Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

Result

No; the positions differ. Repeating one molecule-wide label identifies failure to localise the procedure.

Check

Changed check: the procedure must return allene $sp^2/sp/sp^2$ and propyne $sp^3/sp/sp$ while conserving four carbon basis functions at each site.

Mistake clinic • diagnose and repair

—

A tempting wrong path

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Repair

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- Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: the procedure must return allene $\text{sp}^2/\text{sp}/\text{sp}^2$ and propyne $\text{sp}^3/\text{sp}/\text{sp}$ while conserving four carbon basis functions at each site.

2 • APPLY

State the evidence that would require revisiting a localized lone-pair assignment.

3 • CONNECT

Local count, residual p directions and model applicability have different jobs.

Repair for Q5**Test the rule on a changed case**

Apply the proposed equality-implies-sp² rule to CH₄ and CO₂, then compare it with a domain inventory.

Your prediction and reason

Repair for Q6**Test the rule on a changed case**

Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?

Your prediction and reason

Repair for Q8**Test the rule on a changed case**

During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.

Your prediction and reason

Repair for Q9

Test the rule on a changed case

Compare $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ with $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$. Does spatial closeness alone repair the missing p pathway?

Your prediction and reason

Repair for Q10

Test the rule on a changed case

Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Your prediction and reason

Now you do one

Number carbons left-to-right in $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$. Identify the ordinary contiguous pi pathway and the carbon excluded from that pathway.

Model answer

C1 is the saturated sp^3 methyl centre. C2-C5 are the contiguous p-bearing centres in this introductory model.

1. C1 is the saturated sp^3 methyl centre. C2-C5 are the contiguous p-bearing centres in this introductory model.

Practice this concept in Core2

Q5

Q6

Q8

Q9

Q10